

22521

12425

03 Hours / 70 Marks

Seat No.

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- Instructions* –
- (1) All Questions are *Compulsory*.
 - (2) Answer each next main Question on a new page.
 - (3) Illustrate your answers with neat sketches wherever necessary.
 - (4) Figures to the right indicate full marks.
 - (5) Assume suitable data, if necessary.
 - (6) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

- 1. Attempt any FIVE of the following:** **10**
- a) Write any two benefits of distributed database system.
 - b) Give syntax for Nested query.
 - c) State types of NOSQL database.
 - d) Enlist characteristics of data warehouse.
 - e) State any four components of Hadoop.
 - f) Explain multiset types in SQL.
 - g) Name any four common Mongo DB data types.
- 2. Attempt any THREE of the following:** **12**
- a) Explain Two-tier client server database model with diagram.
 - b) Compare SQL and NO-SQL database system.
 - c) Explain any two basic operation with Mongo DB shell.
 - d) Explain XML document schema.

P.T.O.

- 3. Attempt any THREE of the following:** **12**
- a) Draw and explain data warehousing life cycle.
 - b) Write output for the following command executed on Mongo DB shell.
 - i) > New date ("2023/7/9"/;
 - ii) > "Welcome, MSBTE". replace ("MSBTE", "MongoDB");
 - c) Explain any four applications of data mining.
 - d) Compare between HBase and Hive (any four)
- 4. Attempt any THREE of the following:** **12**
- a) Explain Locked Based Protocols.
 - b) What is the use of find () in MongoDB? Explain it with suitable example.
 - c) Describe any two types of data warehouse architecture.
 - d) Compare data mining and data warehouse.
 - e) Describe Hadoop architecture with diagram.
- 5. Attempt any TWO of the following:** **12**
- a) Explain object identify and reference types in SQL with example.
 - b) Write query to execute find () function on : Inventory.
 - i) To display all documents where status equals either "A" or "G"
 - ii) To display all documents in collection.
 - iii) To display all documents where quality is less than 40 and greater than 20.
 - c) Explain arrays and multiset types in sql with example.

6. Attempt any TWO of the following:**12**

- a) Compare between Parallel and Distributed Database (any six points)
 - b) Explain flower expressions and nested queries in Xquery.
 - c) Explain steps used to perform data analysis in R programming.
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